What momentum says about impact

antocciphysics | TEACHER KEY | CPA

Learning target: Connect stopping impulse to average force without claiming a complete damage forecast.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 3, Day 9; 3 course days later.*
They exchange velocities: A=-3 m/s; B=+5 m/s.

02. Explain a worked example (worked)
Momentum arrows equal; lighter object's velocity twice as long.

03. Look and explain (model)
The .1 s stop has 10 times the average force.

04. Build the solution (practice)
40 kg·m/s; 20 N.

05. Use the idea (reason)
Need energy deposition, trajectory, atmosphere, contact/deformation area, material response and geography; simplistic stopping time may be unjustified.

06. Choose a method (mixed) *Review: Unit 2, Day 16; 12 course days later.*
g=4 N/kg; force 32 N inward.

07. Exit check (exit)
Both double.

08. STRETCH: Apply it another way (stretch)
Each magnitude 24 N average; contact histories and material response may differ.

What momentum says about impact

antocciphysics | TEACHER KEY | HONORS

Learning target: Connect stopping impulse to average force without claiming a complete damage forecast.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 3, Day 9; 3 course days later.*
v_f=1 m/s; K_i=51 J, K_f=3 J. K decreases by 48 J; perfectly inelastic.

02. Explain a worked example (worked)
Momentum arrows equal; lighter object's velocity twice as long.

03. Represent and interpret (model)
The .1 s stop has 10 times the average force.

04. Solve and justify (practice)
Impulse 2×10^8 N·s; force 4×10^8 N; 5 s stop gives 4×10^7 N.

05. Analyze the reasoning (reason)
Need energy deposition, trajectory, atmosphere, contact/deformation area, material response and geography; simplistic stopping time may be unjustified.

06. Choose a method (mixed) *Review: Unit 2, Day 16; 12 course days later.*
v=6 m/s; a≈2.25 m/s²; T≈16.76 s. Speed constant, direction changing.

07. Exit check (exit)
A speed half; $KA=p^2/(2\ mA)=KB/2$.

08. STRETCH: Apply it another way (stretch)
Stop: -5 N·s, -125 N. Rebound: -8 N·s, -200 N; direction reversal increases momentum change.